

Program : Diploma in Wood and Paper Technology	
Course Code : 5129	Course Title: Environmental and Adhesive Technology Lab
Semester : 5	Credits: 1.5
Course Category: Program Elective	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To impart knowledge on the testing of resins
- To familiarize water analysis and predict water quality
- To provide knowledge on monitoring air quality and analyse effluent from paper mills

Course Prerequisites:

Topic	Course code	Course name	Semester
Water pollution		Environmental Science	2
Water treatment, polymers		Applied chemistry	1

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Analyse Resin	12	Applying
CO2	Analyse the water quality	11	Applying
CO3	Analyse the Effluent	12	Applying
CO4	Conduct air quality monitoring	7	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			
CO2		3		3			
CO3				3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Analyse Resin		
M1.01	Laboratory preparation of UF, PF resins	6	Applying
M1.02	To determine the 1.solid content 2. water tolerance 3.pH of resin	3	Applying
M1.03	Estimation of phenol & formaldehyde	3	Applying
CO2	Analyse the water quality		
M2.01	Deduce the pH value of water by pH meter and turbidity in a given water sample	3	Applying
M2.02	Determination of hardness of water(EDTA method)	3	Applying
M2.03	Determination of acidity of water	3	Applying
M2.04	Determination of alkalinity of water	1	Applying
M2.05	Determination of chlorine	1	Applying
	Lab Exam – I	1	
CO3	Analyse the Effluent		
M3.01	Determination of Total solids, dissolved solids & suspended solids	3	Applying
M3.02	Determination of Dissolved oxygen	3	Applying
M3.03	Determination of BOD	3	Applying
M 3.04	Determination of COD	3	Applying

CO4	Conduct air quality monitoring		
M4.01	Estimate the amount of particulate and gaseous matter in ambient air	7	Applying
	Lab Exam – II	2	

Text / Reference:

T/R	Book Title/Author
T1	Tchobanoglous G., Burton B.L., Metcalf L., and Stensel H.D., “Waste Water Engineering” 4th Ed., McGraw Hill 2008
R1	Eckenfelder W.W. and Ford D., “Water Pollution Control” 3rd Ed., Jonkins Publishing Company.2000

Online Resources:

Sl.No	Website Link
1	
2	
3	